

PRIMARY 5 SCIENCE TAO NAN P5 SCIENCE EOY 2025

BOOKLET A AND BOOKLET B

41 questions · Source: Tao Nan School 2025

INSTRUCTIONS TO CANDIDATES

1. Do not turn over this page until you are told to do so.
2. Follow all instructions carefully.
3. Answer **ALL** questions.
4. Write your answers in the spaces provided.

NAME: _____

CLASS: PRIMARY 5 _____ DATE: _____

BOOKLET	TOTAL MARKS	MARKS OBTAINED
A	60	
B	35	
TOTAL	95	

SECTION A: Multiple-Choice Questions

For each of questions 1 to 30, four options are given. One of them is the correct answer. Make your choice (1, 2, 3 or 4) and write the correct number in the brackets provided. (60 marks)

1. The table shows some information on three different animals, A, B and C (a tick shows that the animal has the characteristic):

A: has wings; breathes through lungs.

B: has dry and scaly skin as body covering; breathes through lungs.

C: gives birth to young alive; breathes through lungs.

Which of the following correctly identifies the groups the animals belong to?

(1) A bird, B reptile, C mammal

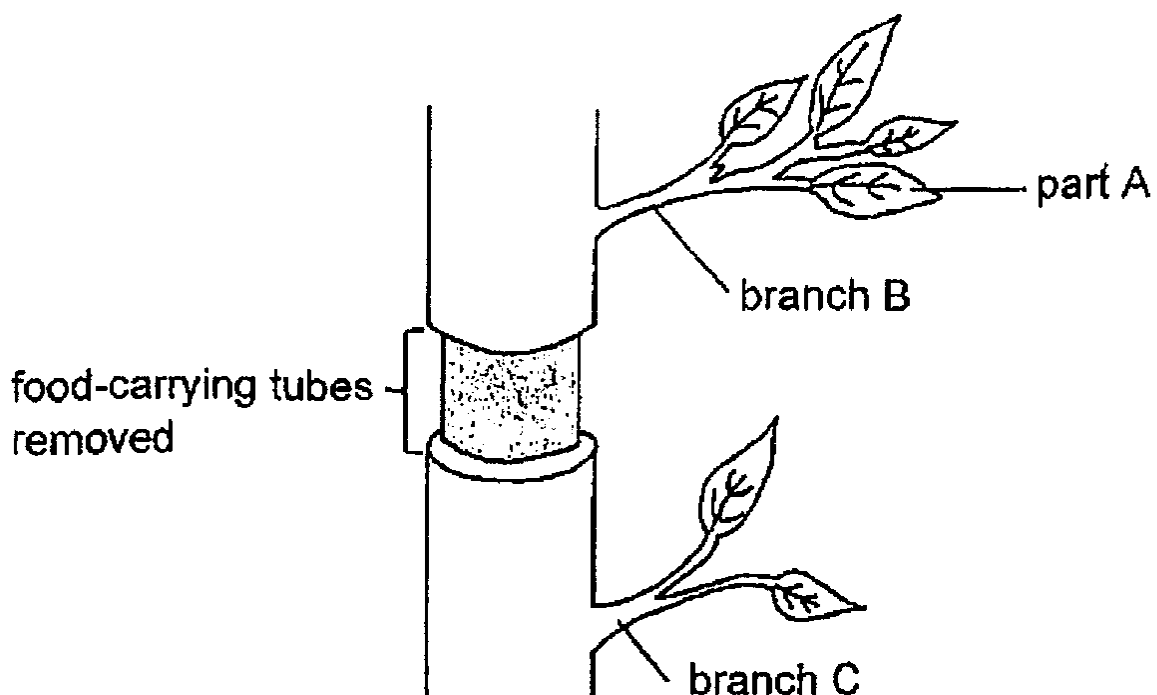
(2) A bird, B fish, C mammal

(3) A reptile, B mammal, C fish

(4) A mammal, B fish, C reptile

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2. Refer to the diagram. Which of the following is the function of part A of a plant?



(1) It supports the plant.

(2) It makes food for the plant.

(3) It takes in water for the plant.

(4) It holds the plant firmly to the ground.

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3. Refer to the same diagram. A farmer removed the food-carrying tubes of the plant (between branch B and branch C). The plant was watered regularly and placed under the Sun. Which of the statements is/are most likely true of the plant after one week?

A: The leaves on branch C died.

B: There was swelling of the stem above and below the cut.

C: The leaves on branch B grew slower than those on branch C.

(1) A only

(2) B only

(3) A and C only

(4) B and C only

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4. Which material is most suitable for making the bouncing net that children can jump on safely?

Material A: flexible ✓, strong ✓, allows light to pass through ✓

Material B: flexible ✓, strong ✗, allows light ✓

Material C: flexible ✓, strong ✓, allows light ✗

Material D: flexible ✗, strong ✓, allows light ✗



(1) A

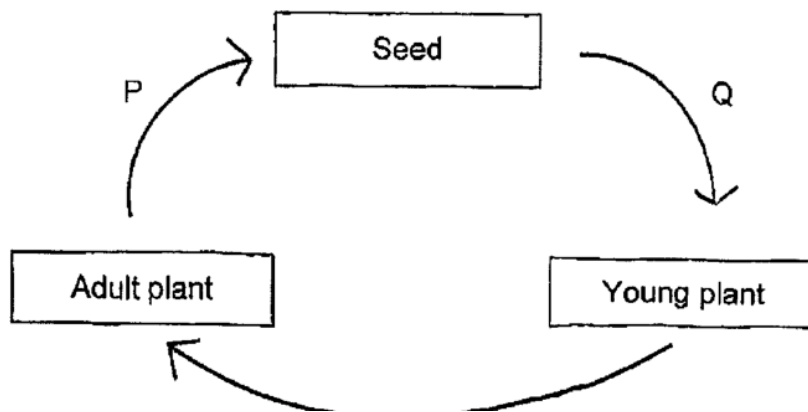
(2) B

(3) C

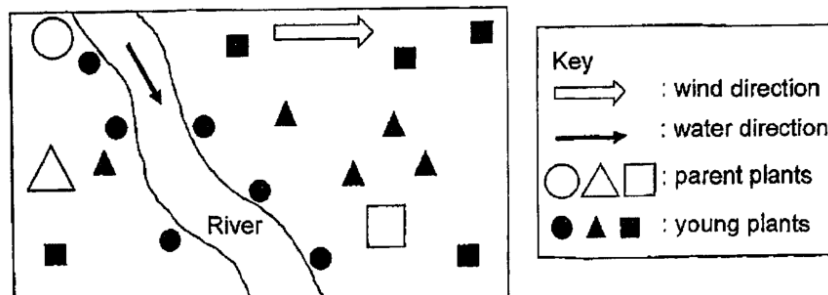
(4) D

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5. The diagram shows the life cycle of a flowering plant: Adult plant → (P) → Seed → (Q) → Young plant → Adult plant. Which of the following processes take place at P and Q?

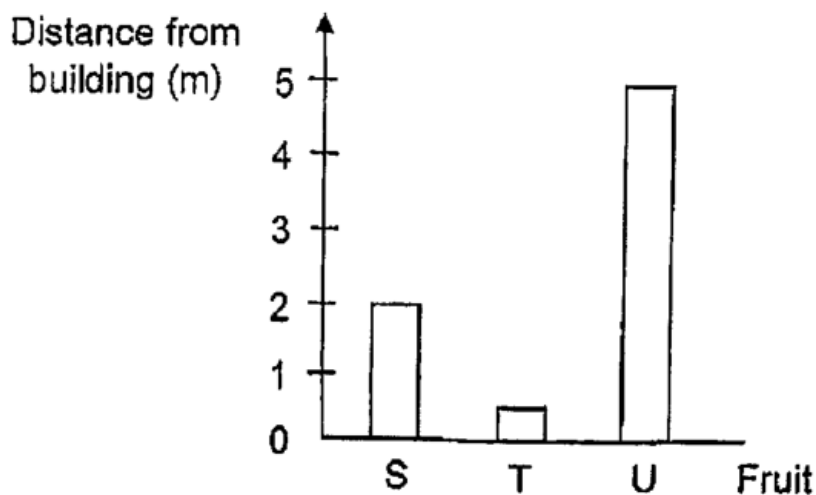


- (1) P: germination, pollination; Q: seed dispersal, fertilisation
 (2) P: pollination, fertilisation; Q: germination, seed dispersal
 (3) P: fertilisation, seed dispersal; Q: pollination, germination
 (4) P: germination; Q: pollination, fertilisation, seed dispersal ()
6. The table shows the characteristics of fruits X, Y and Z: X has a wing-like structure; Y has a fibrous husk; Z is fleshy and juicy. The diagram shows the dispersal patterns of the fruits from parent plants shaped as a circle, a triangle and a square. Which option correctly matches the parent plants (circle, triangle, square) to fruits X, Y and Z?



- (1) circle X, triangle Y, square Z
 (2) circle Y, triangle Z, square X
 (3) circle Z, triangle X, square Y
 (4) circle Y, triangle X, square Z ()

7. Three different fruits, S, T and U, were dropped from the top of a building at the same time. The distance each fruit landed away from the building was measured (see graph). Which of the statements can be concluded from the results of the experiment?
- A: Fruit S has less mass than fruit U.
 B: Fruit T has the largest wing-like structure.
 C: Fruit U is the lightest and the smallest.



- (1) A only
 (2) C only
 (3) A and B only
 (4) B and C only ()
8. Seeds on damp cotton wool were watered daily but placed in different locations: A near the window, B in the freezer, C in a dark cupboard. In which set-up(s) would the seeds most likely germinate?
- (1) A only (2) B only
 (3) A and C only (4) A, B and C ()
9. The table compares parts of the plant and human reproductive systems that have similar functions: plant U ↔ human testes; plant ovary ↔ human V. What could U and V be?
- (1) U anther, V ovule (2) U stigma, V ovary
 (3) U anther, V ovary (4) U stigma, V ovule ()

10. The table shows some characteristics of Amanda and her family:

Amanda: detached earlobes ✓, short hair ✓, brown eyes ✓

Sister: detached earlobes ✓, short hair ✗, brown eyes ✓

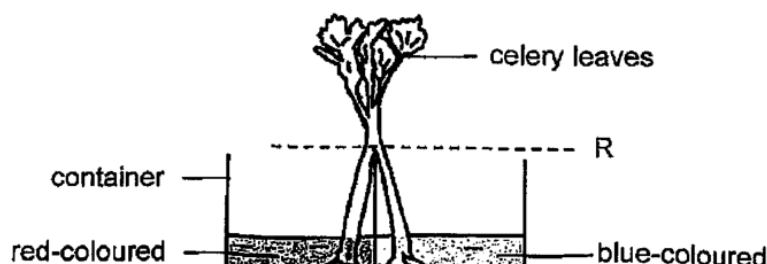
Father: detached earlobes ✗, short hair ✓, brown eyes ✓

Mother: detached earlobes ✓, short hair ✓, brown eyes ✗

Based on the information in the table, which statement is most likely true?

- (1) Amanda inherited 2 characteristics from her father.
- (2) Amanda inherited her detached earlobes trait from her mother and sister.
- (3) Amanda's mother passed down her characteristic of short hair to Amanda.
- (4) Amanda's father passed down his characteristic of brown eyes to Amanda. ()

11. A celery plant was placed into a container as shown. The lower part of the stem was split into two at R; one half stands in red-coloured water and the other in blue-coloured water. Which of the following is a correct observation after one day?



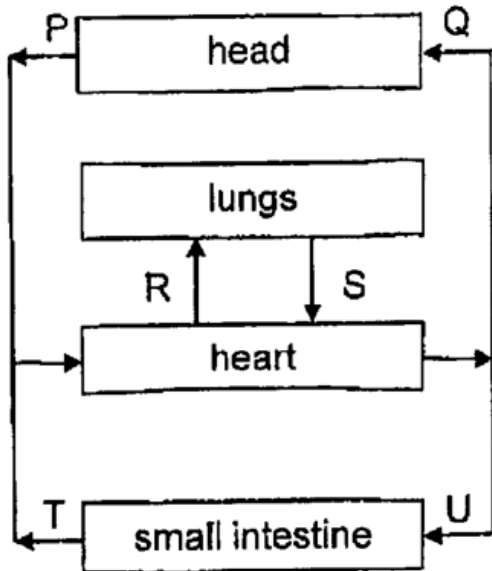
- (1) The leaves remained green.
- (2) All the leaves turned purple.
- (3) The part of the stem above cut R turned purple.
- (4) Some leaves turned red while some leaves turned blue. ()

12. The diagram shows the direction of blood flow in certain parts of the body a few hours after a meal. Which statement(s) is/are correct?

A: Blood in P has less oxygen than blood in Q.

B: Blood in R has less carbon dioxide than blood in S.

C: Blood in T has less digested food than blood in U.



- (1) A only
- (2) C only
- (3) A and B only
- (4) B and C only

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13. A boy is blowing into a balloon. Compare the gases in the surrounding and in the balloon. Which statement is correct?

- (1) There is less oxygen in the surrounding than in the balloon.
- (2) There is less water vapour in the surrounding than in the balloon.
- (3) There is more carbon dioxide in the surrounding than in the balloon.
- (4) There is no change in the percentage of gases in the surrounding and in the balloon.

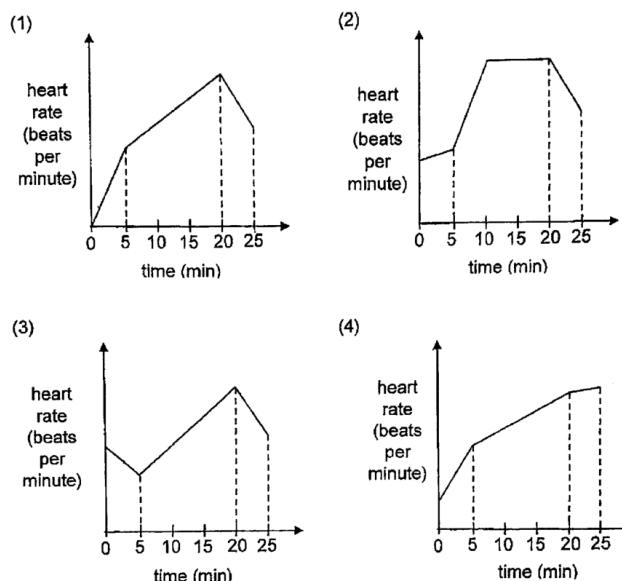
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14. Which part is involved in the exchange of gases in a plant, fish and human?

- (1) plant: leaves; fish: mouth; human: lungs
- (2) plant: leaves; fish: gills; human: nose
- (3) plant: tiny openings; fish: gills; human: lungs
- (4) plant: tiny openings; fish: mouth; human: nose

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15. Alvin walked for 5 minutes to the park. At the park, he cycled for 15 minutes before resting for 5 minutes. Which of the graphs (1)–(4) correctly shows his heart rate?



(1) (1)

(2) (2)

(3) (3)

(4) (4)

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16. The diagram shows a kettle of boiling water with a 'white cloud' near the spout. What is the 'white cloud' that can be seen?

(1) steam

(2) smoke

(3) water vapour

(4) water droplets

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17. Substance A is a liquid at 21 °C and a gas at 92 °C. Which of the following are possible freezing and boiling points of substance A?

(1) freezing 0 °C, boiling 100 °C

(2) freezing 6 °C, boiling 80 °C

(3) freezing 26 °C, boiling 161 °C

(4) freezing 41 °C, boiling 182 °C

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18. Which one of the following statements about boiling and melting is INCORRECT?

(1) Both processes involve a change of state.

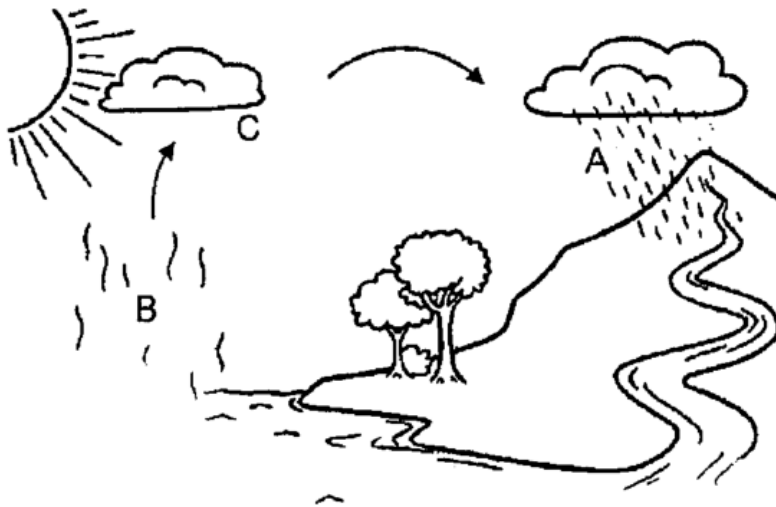
(2) Both processes occur at a fixed temperature.

(3) The boiling point is always higher than the melting point.

(4) One process involves heat gain and the other process involves heat loss.

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19. The diagram represents the water cycle. A, B and C are processes that occur in the water cycle. Which of the following about the processes in the water cycle is correct?



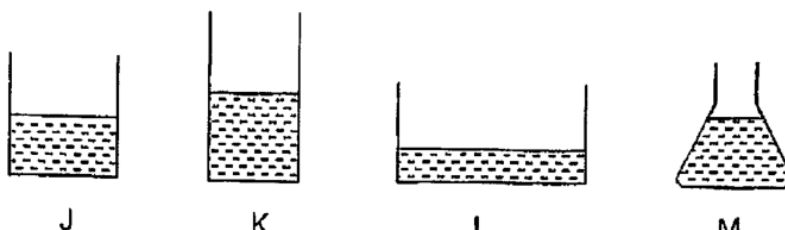
- (1) A — heat loss
- (2) B — heat gain
- (3) B and C — heat gain
- (4) C and A — no heat gain or heat loss ()

20. Which of the following definitely helps to conserve water?

A: repair a leaking tap
 B: use a mug when brushing teeth
 C: take a longer shower
 D: use the water hose when watering the plants

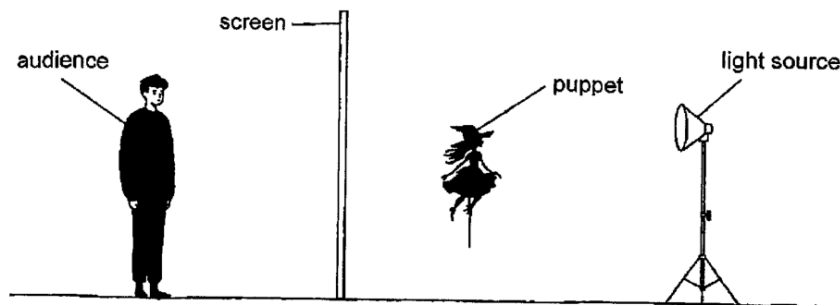
- (1) A and B only
- (2) A and C only
- (3) B and C only
- (4) C and D only ()

21. Four containers J, K, L and M hold the same amount of water. They are left near an open window for a few hours. Arrange the containers from the least volume of water left after a few hours.



- (1) J, L, K, M
- (2) M, K, J, L
- (3) L, J, K, M
- (4) K, J, M, L ()

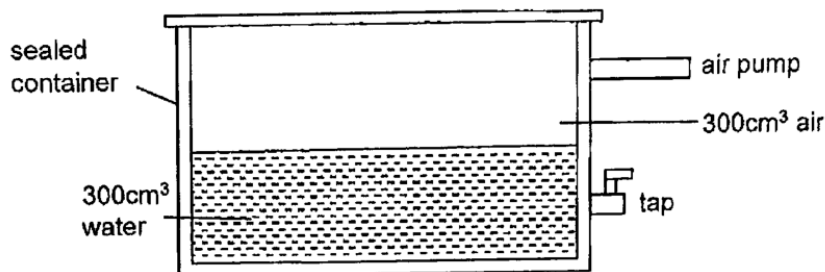
22. Jerry arranged a set-up (audience | screen | puppet | light source) to perform a shadow puppet show. Which of the following explains how the shadows are formed so that the audience can enjoy the show?



- (1) puppet allows no light to pass through; screen allows some light to pass through
- (2) puppet allows most light to pass through; screen allows no light to pass through
- (3) puppet allows no light to pass through; screen allows most light to pass through
- (4) puppet allows some light to pass through; screen allows no light to pass through

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23. A sealed container holds 300 cm^3 of air and 300 cm^3 of water. A volume of 100 cm^3 of air was removed from the container through the air pump while 150 cm^3 of water was removed through the tap. What was the final volume of the air in the container?



- (1) 150 cm^3
- (2) 200 cm^3
- (3) 300 cm^3
- (4) 450 cm^3

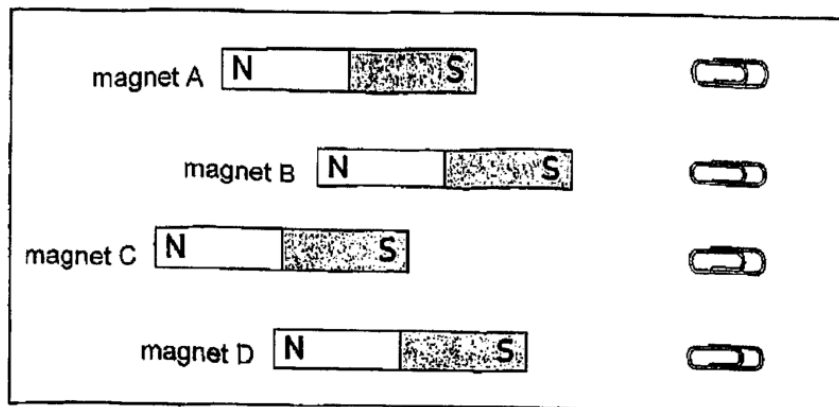
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24. When boiling water was poured into a thick glass cup that was just taken out of the freezer, the walls of the glass cracked. X is the inner part of the glass and Y is the outer part of the glass. What would be the best explanation why the glass cracked?

- (1) X gained heat but Y did not.
- (2) X gained heat but Y lost heat.
- (3) X lost more heat than Y and contracted faster.
- (4) X gained more heat than Y and expanded faster.

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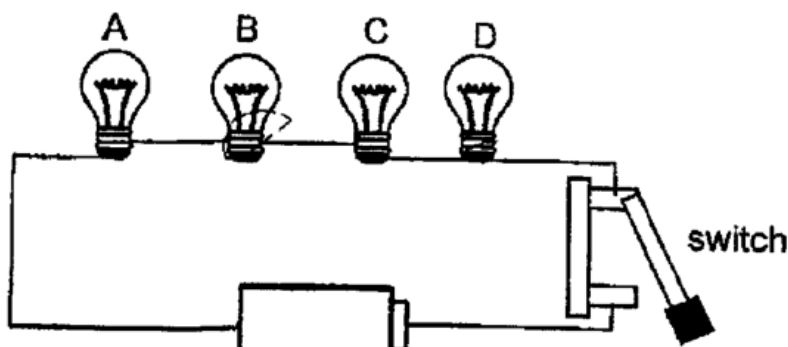
25. The diagram shows the positions at which each magnet A, B, C and D started to attract the paper clip. Which magnet was the strongest?



- (1) Magnet A
- (2) Magnet B
- (3) Magnet C
- (4) Magnet D

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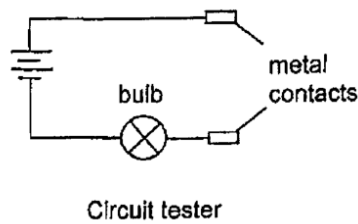
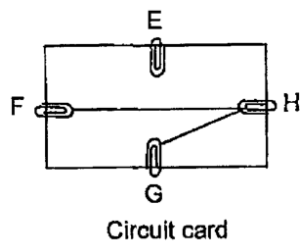
26. Billy set up the circuit shown. Which bulb(s) will light up when the switch is closed?



- (1) A and C only
- (2) B and D only
- (3) A, B and C only
- (4) None will light up.

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27. Study the circuit card (clips E, F, G, H) which is tested with the circuit tester. Which set of results is INCORRECT?

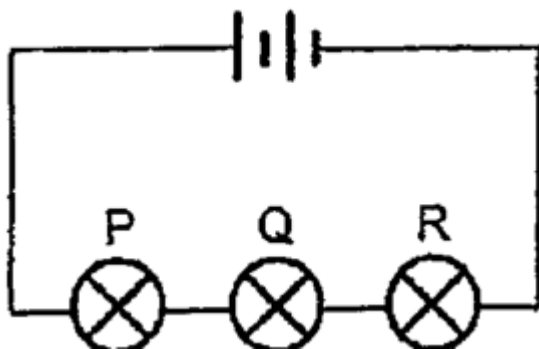


- (1) E and F — does not light up
- (2) F and G — does not light up
- (3) E and G — does not light up
- (4) F and H — lights up

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28. Sanjay set up a circuit with bulbs P, Q and R in series. Which of the following should Sanjay do to increase the brightness of bulb Q?

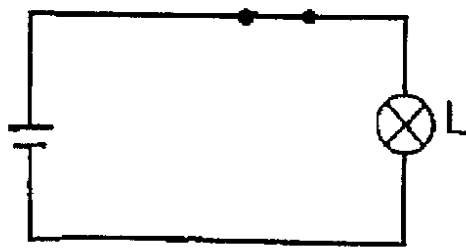
- A: Remove bulb R.
- B: Remove one battery.
- C: Add one more battery.
- D: Rearrange all the bulbs to be in parallel to one another.



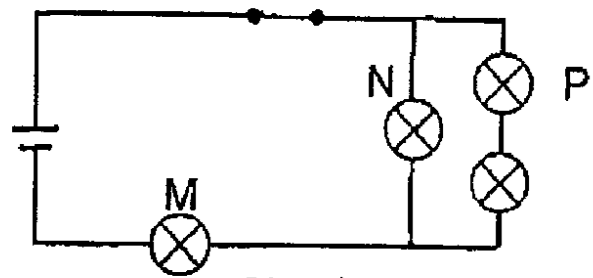
- (1) A and C only
- (2) B and D only
- (3) A, C and D only
- (4) A, B and D only

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- 29.** Two circuits are set up as shown. All bulbs are identical and the components are in working condition. Which bulb in Circuit 2 will have the same brightness as bulb L in Circuit 1?



Circuit 1



Circuit 2

- (1) Bulb M
 - (2) Bulb N
 - (3) Bulb P
 - (4) None of the bulbs in Circuit 2
- ()

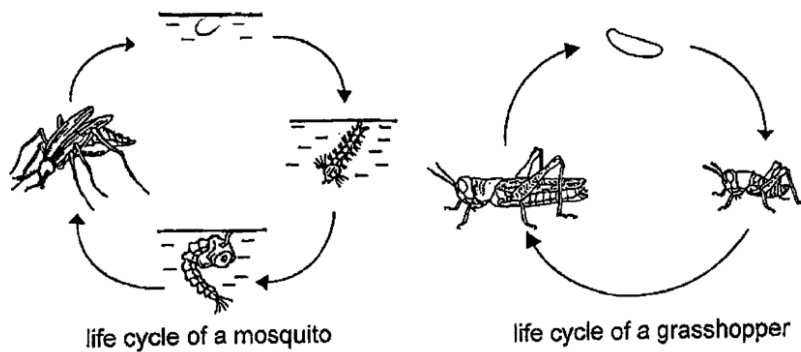
- 30.** (Same two circuits.) What will happen when bulb P is removed from Circuit 2?

- (1) Only bulb M will light up.
 - (2) Only bulb N will light up.
 - (3) Bulbs M and N will light up.
 - (4) None of the bulbs will light up.
- ()

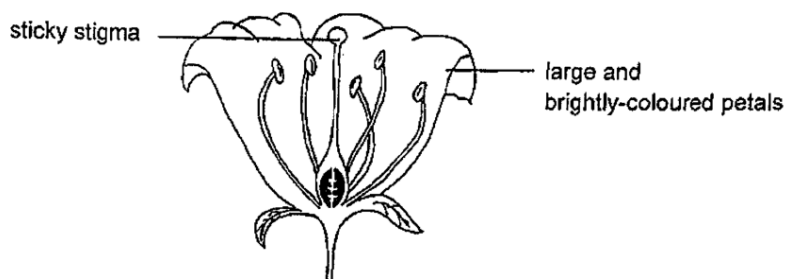
SECTION A: Open-Ended Questions

For questions 31 to 41, write your answers in the spaces provided. (35 marks)

- 31.** The diagrams show the life cycles of a mosquito and a grasshopper. [3]
- (a) Based on the diagrams, state two differences between the life cycles of the mosquito and the grasshopper.
- (b) Name another animal that has the same number of stages as the life cycle of the mosquito.

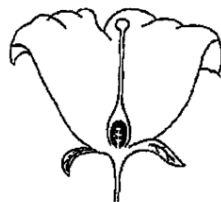


- 32.** The diagram shows a flower that Jenny found in her garden (sticky stigma; large and brightly-coloured petals). [4]
- (a) What is pollination?
- (b) Based on the diagram, what is the likely method of pollination of this flower? Explain your answer.
- (c) Jenny removed a certain part of the flower from each of three flowers, K, L and M (second diagram). After some time, which flower(s) can develop into fruits? Explain why.





K



L



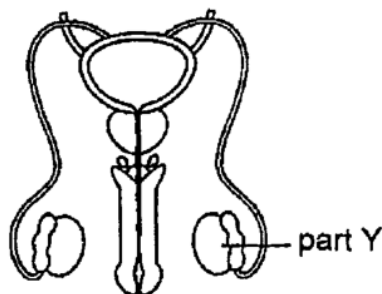
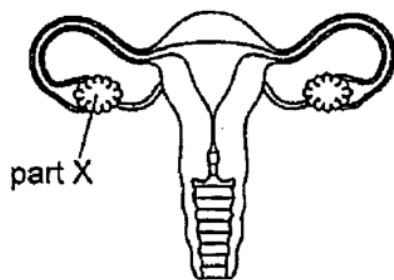
M

33. The diagrams show the female and male human reproductive systems. [3]

(a) Identify part X and part Y.

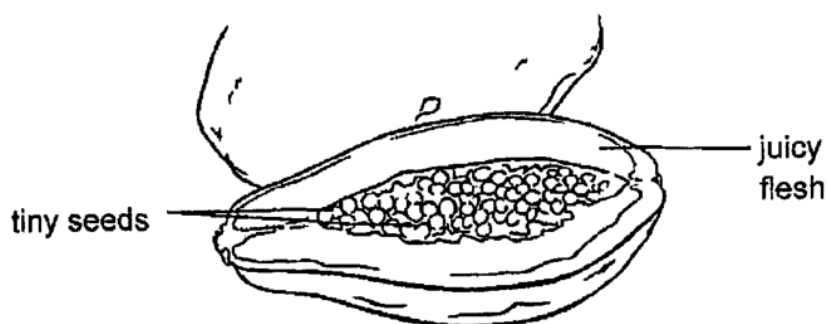
(b) State what fertilisation is.

(c) Due to health reasons, an adult male needs to have part Y removed. Will he still be able to have an offspring? Explain why.



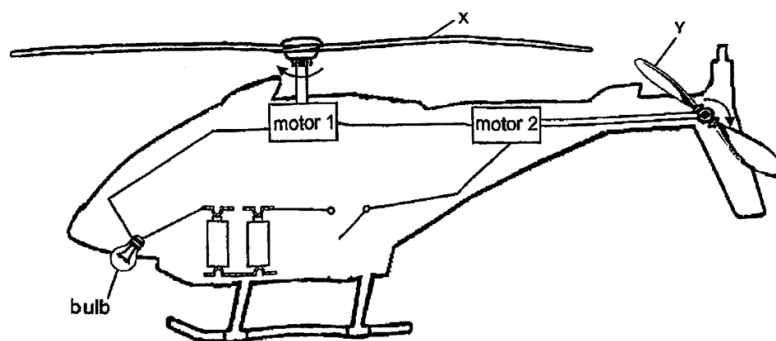
- 34.** Timmy was hiking up a mountain. He wanted to find out how his breathing rate at rest was affected when he went higher up the mountain. He recorded: [5]
- Height 1500 m — oxygen 17% — 16 breaths per minute
Height 2000 m — oxygen 16% — 20 breaths per minute
Height 2500 m — oxygen 15% — ?
Height 3000 m — oxygen 14% — 24 breaths per minute
- (a) Write down a possible reading of Timmy's breathing rate at rest at 2500 m.
(b) What could Timmy conclude about how his breathing rate at rest was affected by the height above sea level?
(c) Based on the data, explain your answer in (b).
(d) State how sitting for 30 minutes before measuring his breathing rate at each height helped to make the investigation a fair test.

- 35.** The diagram shows a fruit with juicy flesh and many tiny seeds. [3]
- (a) State an advantage of this fruit having many seeds.
(b) Describe how the seeds of this fruit can be dispersed over a wide area.

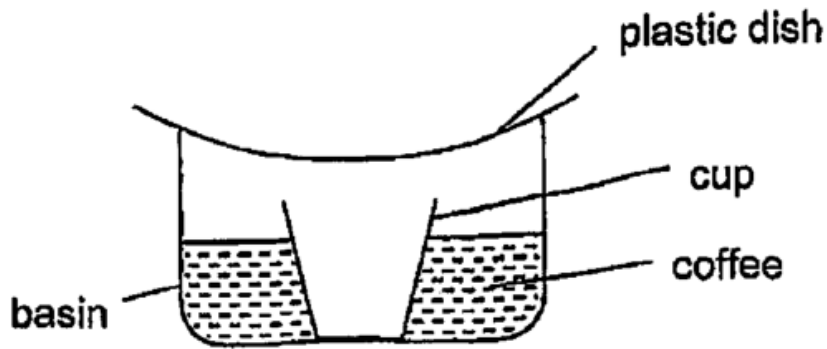


36. Four identical containers, filled with the same volume of water, were left in four different locations for 2 hours: A cloudy, no wind; B cloudy, windy; C sunny, windy; D sunny, no wind. [3]
- (b) What is the aim of this experiment?
- (c) In which of the four locations will wet clothes dry the fastest? Explain why.
- (Part of the original question is not included here.)

37. The diagram shows the circuit inside a remote-controlled toy helicopter. All components are in working condition. When the switch is closed, the bulb should light up while propellers X and Y spin, but the bulb did not light up and neither propeller spun. Motor 1 turns X and motor 2 turns Y if electricity flows through the circuit. [1]
- (a) Based on the diagram, give a reason why the toy is not working.
- (Part of the original question is not included here.)



38. Fred placed a cup into a basin. He poured some coffee into the basin and covered it with a plastic dish. He then placed the basin under the hot sun for a few hours. After a few hours, Fred found substance F in the cup. [3]
- (a) What is substance F?
- (b) State the two processes that took place for substance F to be collected in the cup.
- (c) Without changing the plastic dish, suggest a change to the set-up to increase the amount of substance F in the cup within the same duration.



39. A cup of water at 10 °C (in a metal cup with a lid) was placed in a room at a temperature of 29 °C. [3]

(a) What is the temperature of the water in the cup after a few hours?

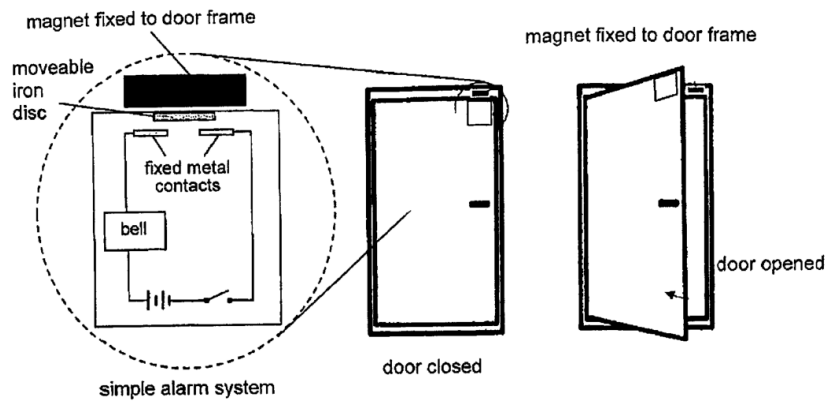
(c) The cup was then placed in the freezer where the temperature is below 0 °C. What is the state of the water in the cup after a few hours? Explain why.

(Part of the original question is not included here.)

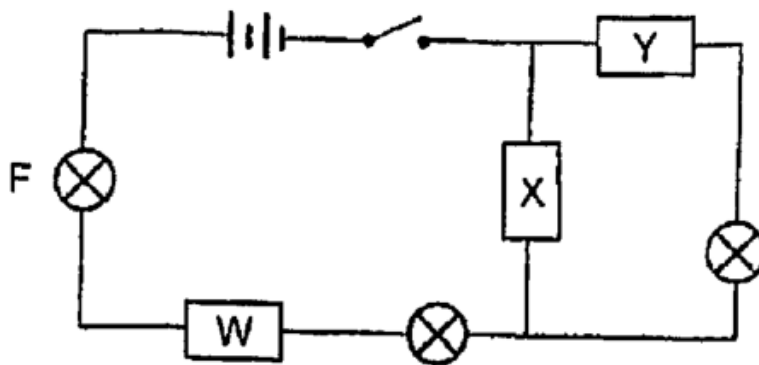
40. Gary made a simple alarm system which he installed onto the door and door frame of his office. If the switch is closed, the bell will sound when the door is opened. [3]

(a) Based on the diagrams, explain how the alarm system works when a burglar opens the door if the switch was closed earlier on.

(b) Gary wanted to add a bulb to his alarm system. How should Gary connect the bulb to the circuit such that he could be warned by the lit bulb even when the bell is not working?



41. A circuit was set up using 3 rods of different materials, W, X and Y. Only one of them is an insulator of electricity. All components are in working condition. [4]
- (a) When the switch was closed, two bulbs lit up. Which rods are conductors of electricity and which is the insulator?
- (b) What would happen to the brightness of bulb F if another bulb is added in series to it? Explain why.
- (c) If rod W was replaced with a plastic rod, how many bulbs would light up? Explain why.



— END OF PAPER —